

$X \rightarrow$ random variable



$\{x_1, x_2, x_3, \dots, x_{10}\} \leftarrow$ sample $\nearrow$ population mean



Task $\rightarrow$ estimate population mean of $X = \mu$



$$\mu = \bar{x} \Rightarrow \frac{1}{n} \sum_{i=1}^n x_i$$

$$\boxed{\mu = 74}$$

$$\mu \Rightarrow [70, 80]$$



' interval

$$\boxed{\mu = \bar{x}} \leftarrow \text{specified}$$

$\rightarrow$ point estimation

$$\rightarrow \boxed{100}$$



prediction

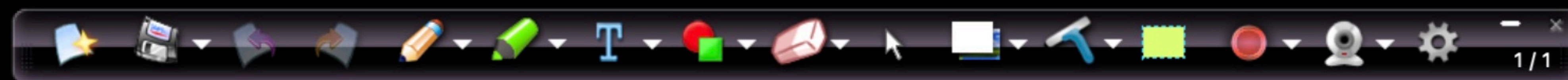
$$[100, 120] \uparrow$$



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You are screen sharing



Stop share

computing C.I given underlying distributions

$$X \sim N(\mu, \sigma)$$

mean = 168 cm

std-dev = 5 cm

||

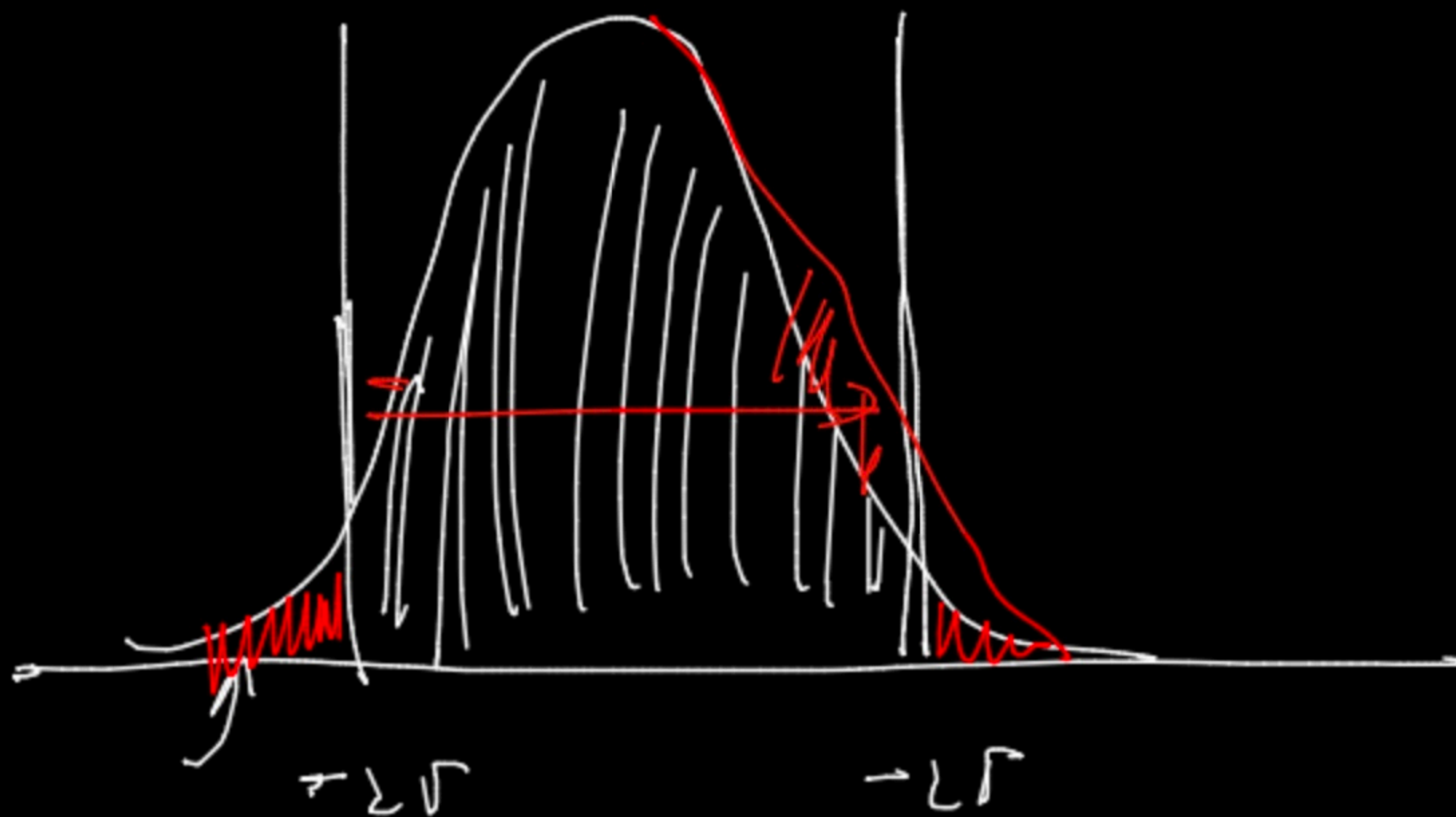
95%

$$\mu - 2\sigma, \mu + 2\sigma$$

[]

↑

$$[158, 178] \rightarrow 95\%$$



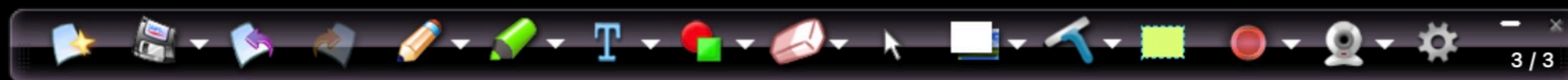
2σ → 95% ^{95%}



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C.I for mean(μ) of a r.v. (x)

$X \sim F \rightarrow \text{mean} \Rightarrow \mu$
 $\searrow$
 std-dev = σ

$\{x_1, x_2, x_3, \dots, x_n\}$

$S_1 \rightarrow \bar{x}_1$

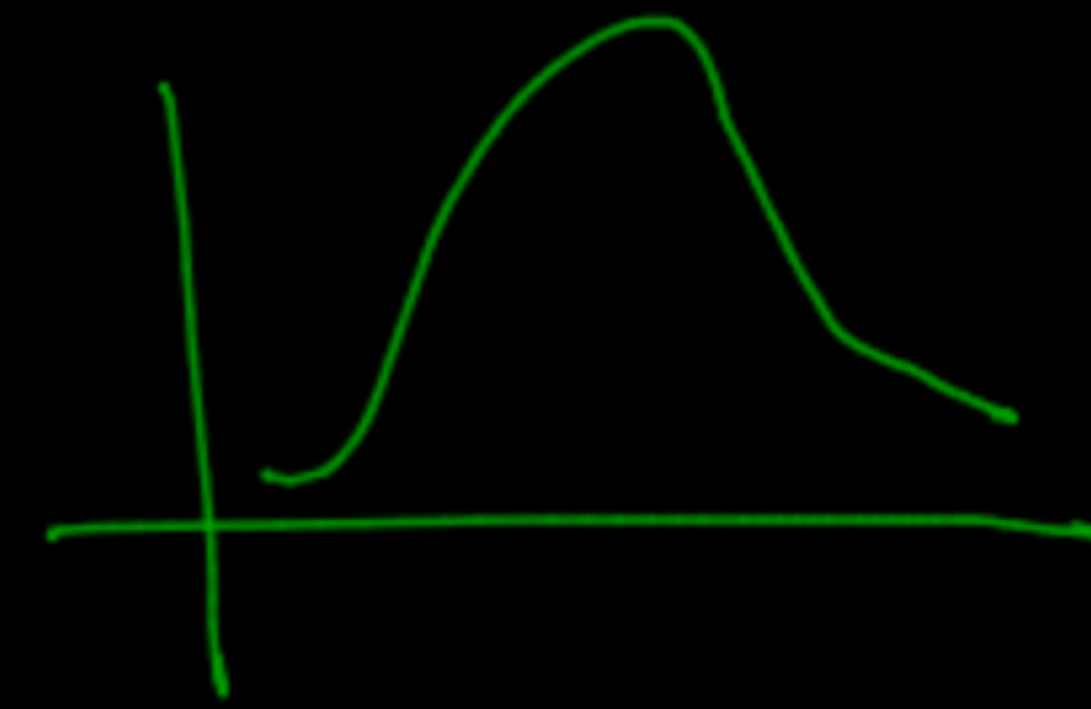
$S_2 \rightarrow \bar{x}_2$

$S_3 \rightarrow \bar{x}_3$

$S_4 \rightarrow \bar{x}_4$

$\vdots$
 $S_n \rightarrow \bar{x}_n$

C.L.T



$\uparrow \text{mean} = \mu_x$

¹⁶⁸
 $\text{sum} \leftarrow \{180, 162, 158, 172, 168, 150, 171, 183, 165, 176\}$

Q what is the 95% C.I $\rightarrow \mu$ (x)

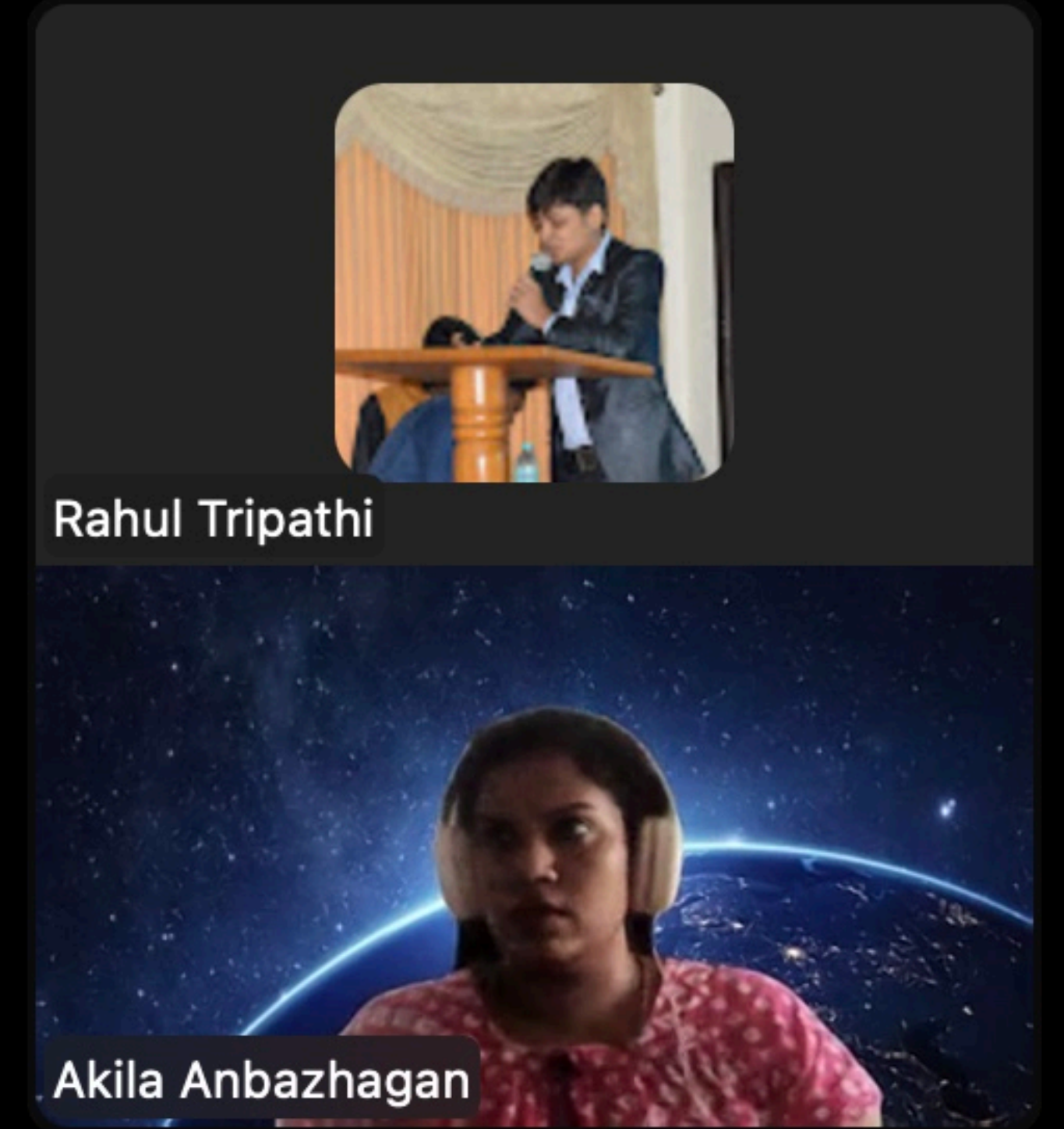
[]

$N(\mu, \sigma)$

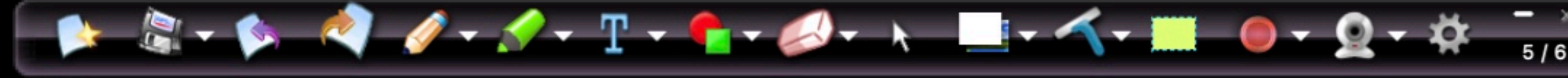
95%
 $\left[\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \right] \leftarrow \mu_x + z^* \frac{\sigma}{\sqrt{n}}$

$[165.34, 171.66] \rightarrow \underline{\text{95\%}}$

$X \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$



Hypothesis testing



$\Rightarrow$ Is there any diffⁿ b/w μ of class A and class B

Class A

160
152

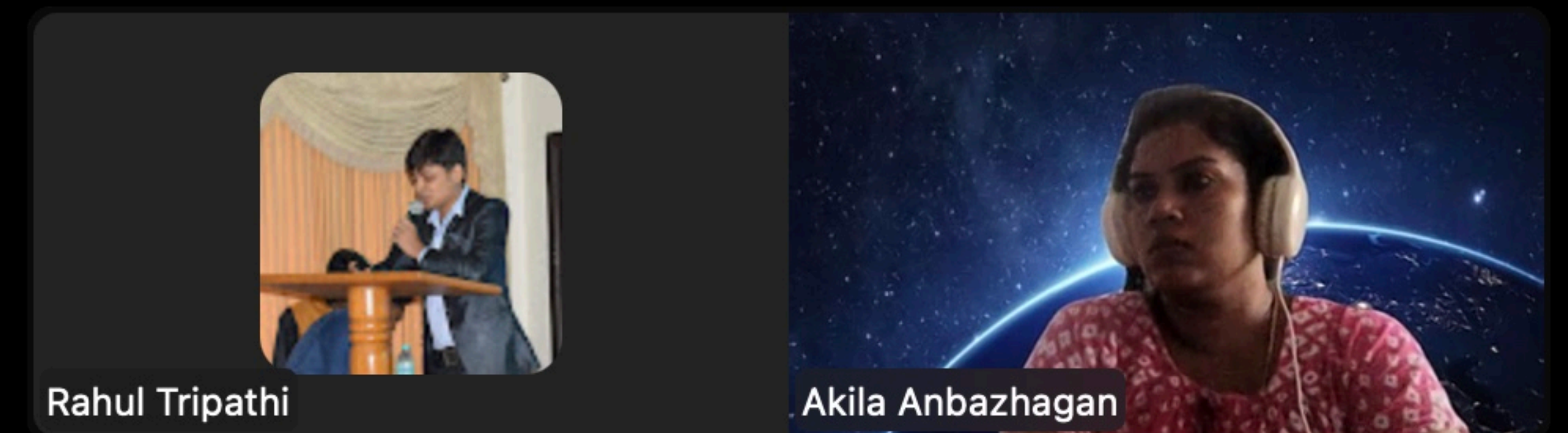
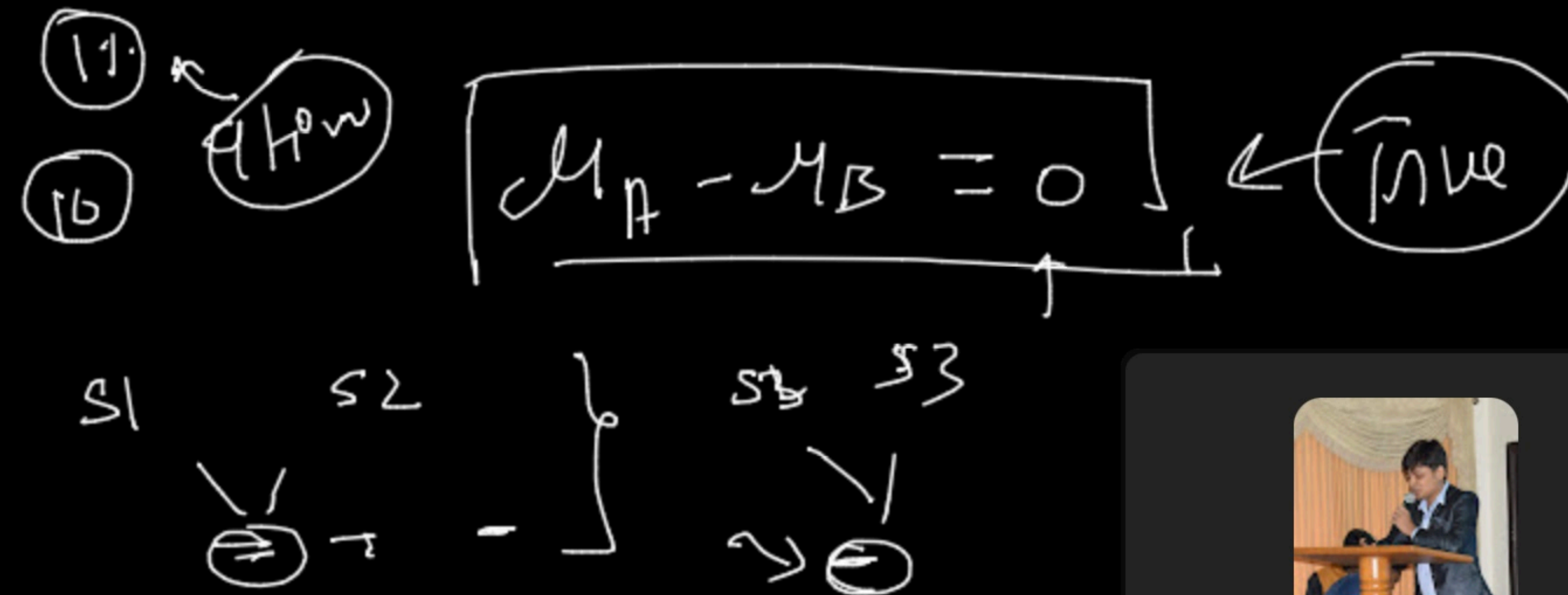
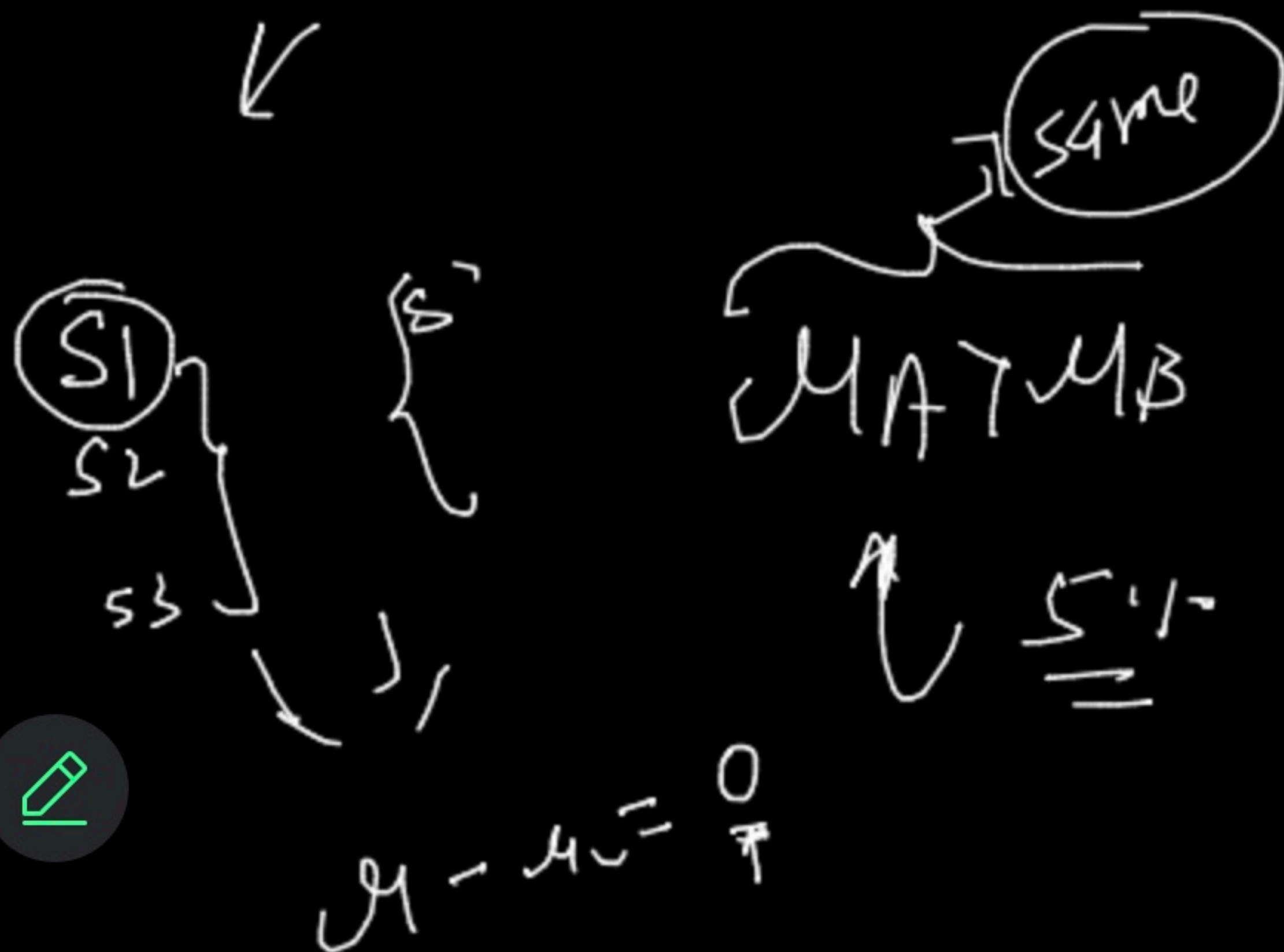
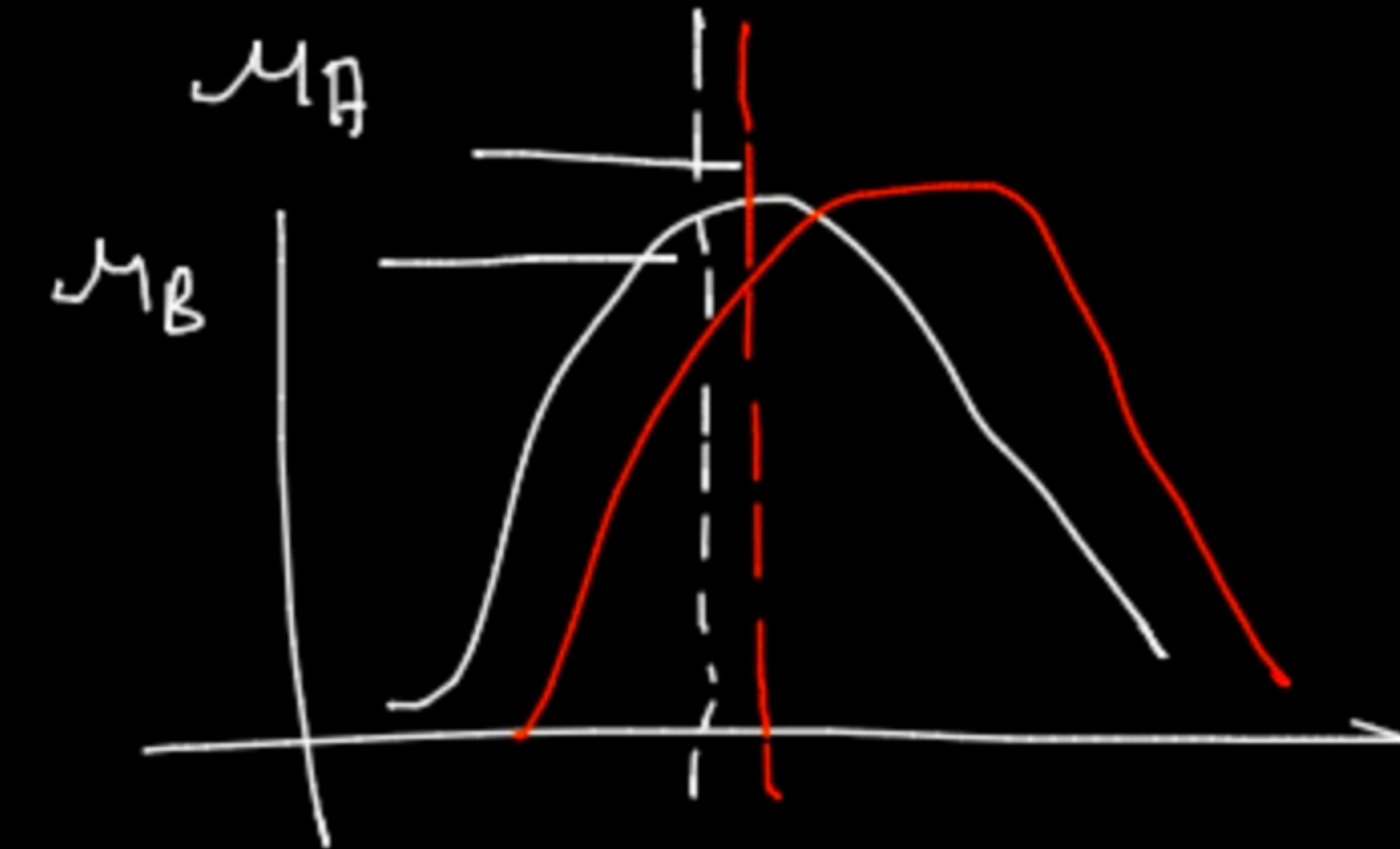
148

Class B

162
158

182

μ_A μ_B



Step 1 Choose test-statistics
↑
 $(\mu_B - \mu_A) \neq 0$ + same Step 3

Step 2 Define NULL Hypothesis (H_0)

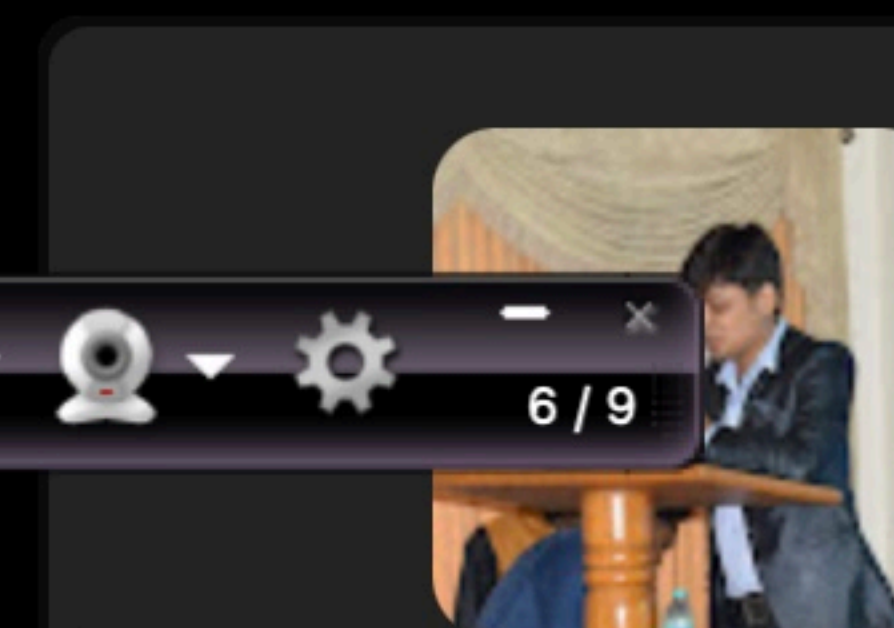
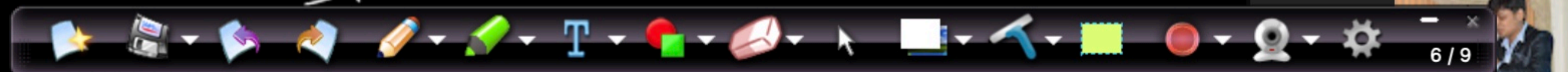
H_0 : make some assumption

↓
 H_1 : Alternative Assumption
inverse of (H_0)

like
 H_0 : no diffⁿ in test $(\mu_B - \mu_A = 0)$

False/true

Accept ✓



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Step 3

Compute

p-value
↑ ↑

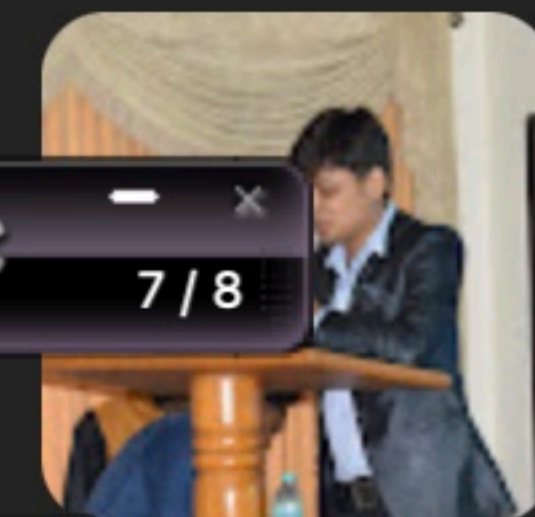
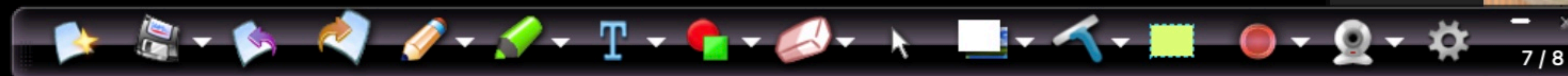
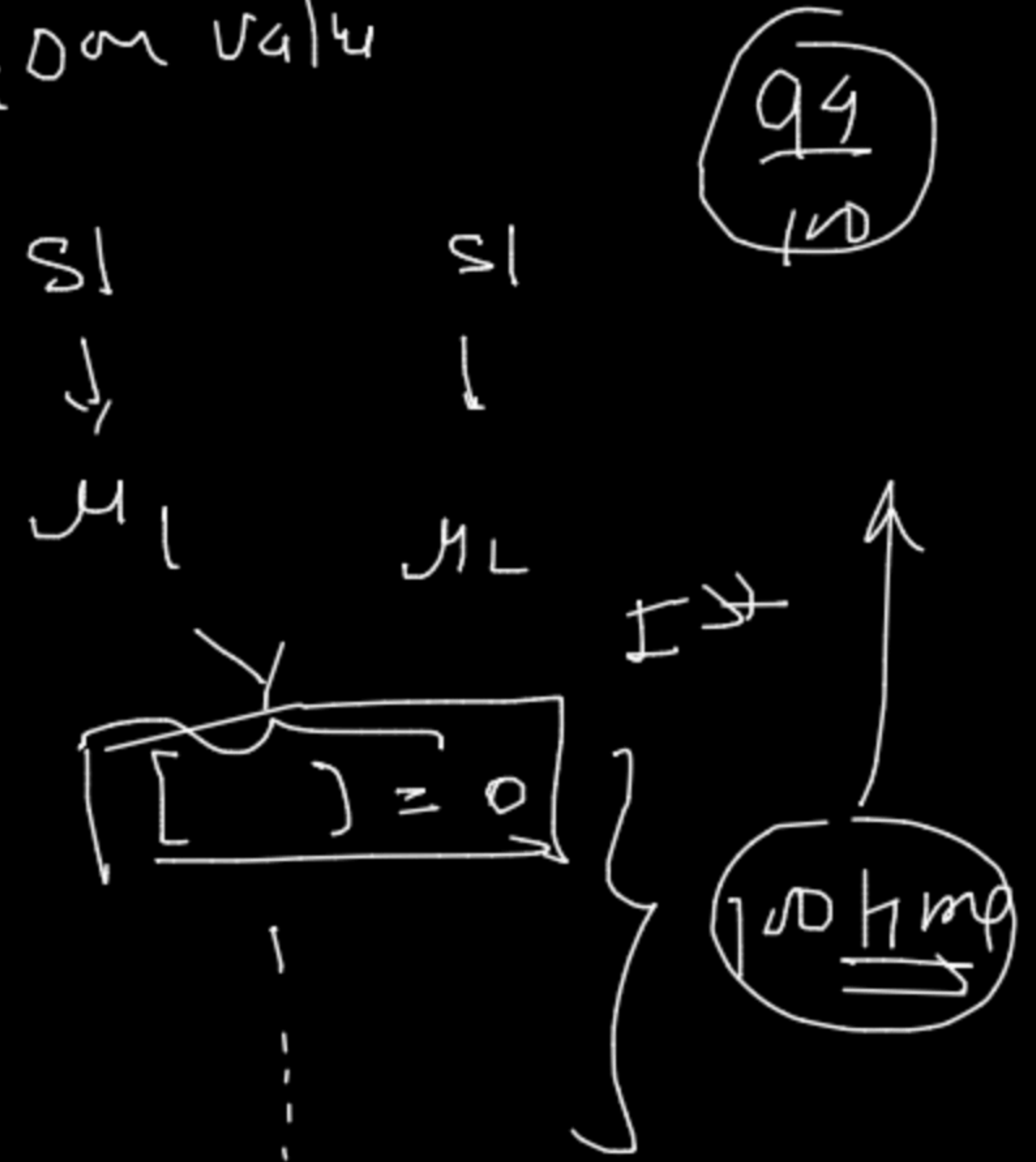
what is the prob of observing your value

ie. $\mu_B - \mu \geq 0$

① $p\text{-value} = 0.9$ It means that
prob of obs. Null
Hypothesis $\boxed{0.9}$

② $p < 0.05$
↑
Reject H_0

if H_0 is True

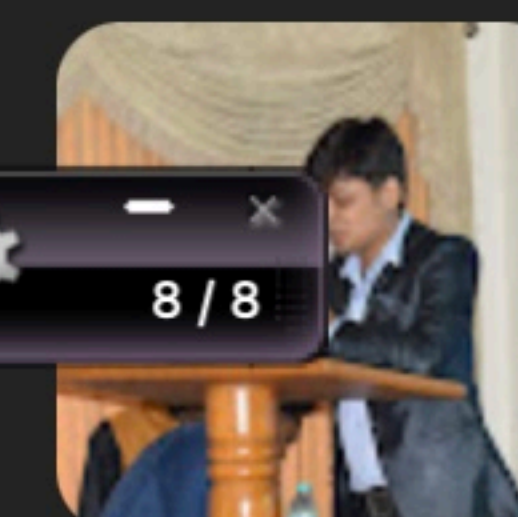


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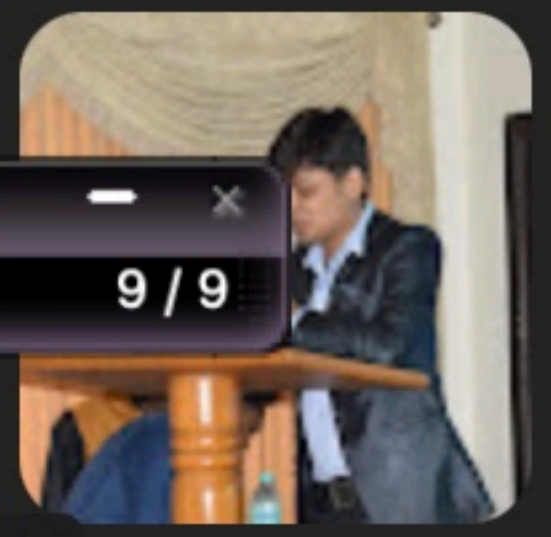
$H_0 \rightarrow \mathcal{H}_B - \mathcal{H}_A = 0$
 $=$
 $H_1 \rightarrow \boxed{14}$ opposite



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